NISAR Water Analysis in Bihar

L-band and S-band GCOV inspection, exploratory fusion, and candidate mapping

AquaFusion project | Technical progress report | 10 November 2026

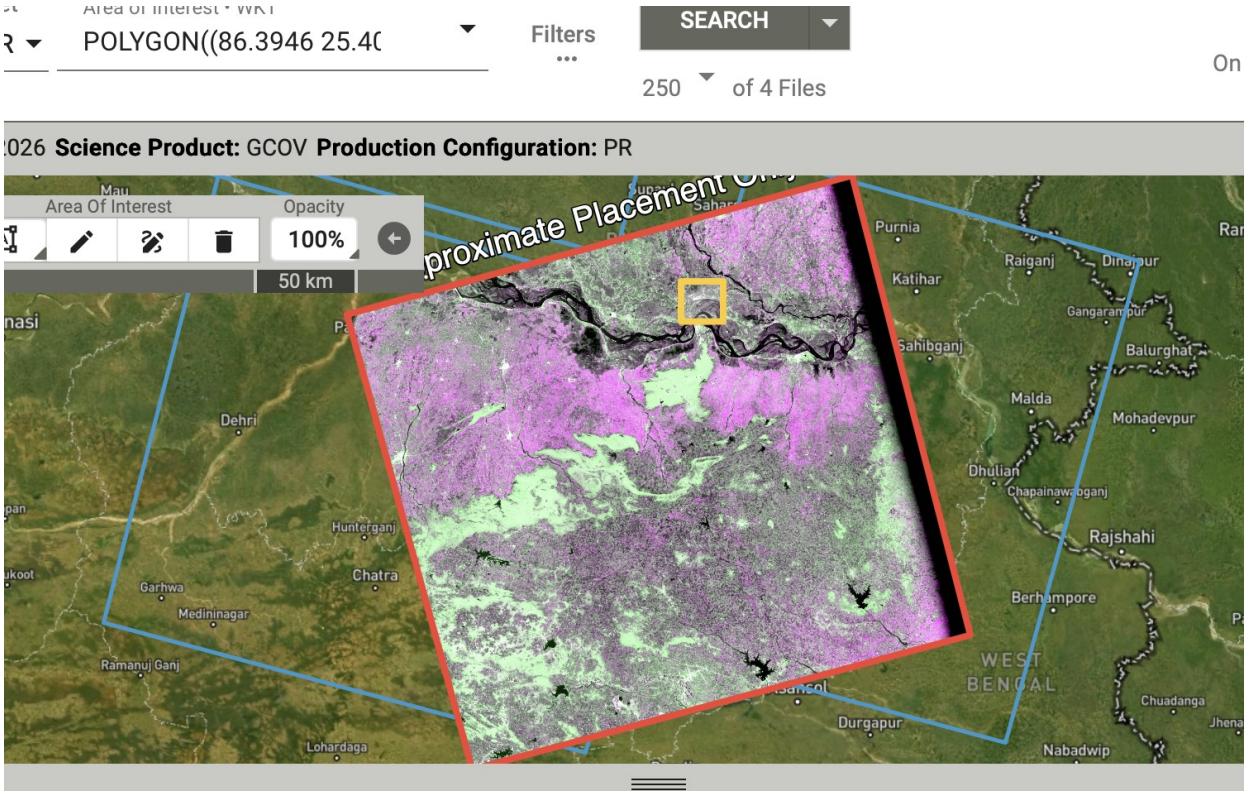
This progress report summarizes the first NISAR L-band and S-band water-screening work for the Bihar study area. Both products cover the selected centre, and their shared 800 × 800 pixel window supports a direct comparison of HH and HV backscatter. Equal-weight and local-contrast weighted low-backscatter scores have been generated, along with threshold and cross-band agreement checks. The maps identify candidate areas for review; they are not yet validated water maps, and this single-date pair cannot measure change through time.

Item	Current evidence
Study reference	25.5000° N, 86.5000° E; supplied 20 km reference box
Acquisition	30 October 2026; Track 98, Frame 14; near-simultaneous L/S observations
Grid	EPSG:32 645; 10 m × 10 m spacing; shared 800 × 800 analysis window
Exploratory outputs	HH/HV maps, equal-weight score, local-contrast weighted score, threshold and agreement diagnostics
Interpretive status	Candidate mapping only; no ground-truth accuracy assessment or temporal change estimate

The key spatial check passed: the supplied centre is inside both grids, and both nearest pixels have mask value 1. The shared window contains 640,000 jointly valid pixels (64 km²). At the 10th-percentile HH cutoffs, 35,651 pixels are selected by both bands; 34,033 also fall in the top decile of the equal-weight score. These areas remain exploratory until validated against independent reference data.

1. Study question and scope

The project investigates surface-water and inundation signals in the Ganga alluvial plain of Bihar using NISAR L-band and S-band SAR. The supplied reference centre is 25.5000° N, 86.5000° E, with a 20 km × 20 km bounding box extending from 86.3946° to 86.6054° E and 25.4098° to 25.5902° N. The selected reference centre is a coverage check; the notebook visualizations shown here use a 800 × 800 pixel subset around the centre rather than a full-scene classification.



Selected NISAR scene footprint over the Bihar study area; the highlighted location marks the reference centre.

The NISAR mission carries L-band and S-band radar instruments, and GCOV products provide calibrated, terrain-corrected, geocoded polarimetric covariance measurements suitable for applications including inundation mapping [1, 2].

2. Inputs and product quality

Metadata	L-band product	S-band product
Product family	NISAR L2 PR GCOV	NISAR L2 UR GCOV
Band / instrument	L-band / LSAR	S-band / SSAR
Acquisition	30 Oct 2026; Track 98, Frame 14	30 Oct 2026; Track 98, Frame 14
Zero-Doppler interval	23:38:11 to 23:38:45 UTC (filename metadata)	23:38:11.056 to 23:38:47.777 UTC
Raster size	33,696 rows × 34,344 columns	33,120 rows × 34,200 columns
CRS	EPSG:32 645	EPSG:32 645
Pixel spacing	+10 m X; −10 m Y	+10 m X; −10 m Y
X coordinate range	244,805 to 588,235 m	245,525 to 587,515 m
Y coordinate range	2,569,685 to 2,906,635 m	2,572,565 to 2,903,755 m
Covariance layers	HHHH, HVHV	HHHH, HVHV, HHHV
Polarizations listed	HH, HV	HH, HV

The S-band QA report identifies an ISRO Level-2 GCOV product, product version 1.2.1, frequency A, ascending pass, left-looking geometry, and a joint observation. Its QA checklist reports passes for product file availability, filename convention, HDF5 content, image content, and empty-field checks. The report gives Sigma0 statistics of −9.575 dB mean and 3.706 dB standard deviation for HH, and −15.842 dB mean and 3.168 dB standard deviation for HV [3]. These are product QA statistics and are not the same sample as the notebook's local-window percentile summaries.

3. Grid alignment and data-quality mask

Both products use EPSG:32 645 (WGS 84 / UTM zone 45N), 10 m pixel spacing, and overlapping coordinate extents. The intersection of the reported coordinate bounds is E 245,525–587,515 m and N 2,572,565–2,903,755 m. The official centre

falls within this overlap. At the nearest sampled grid location, both products report E 449,755 m, N 2,820,405 m and mask value 1.

The GCOV mask is interpreted according to the product definition: values 1–254 identify valid subswath samples, 0 marks samples that are not fully focused/valid, and 255 marks pixels outside the radar acquisition extent [3]. The analysis therefore used `(mask > 0) & (mask != 255)` rather than treating all nonzero values as a generic land/water class. The mask is a data-validity layer, not a water map.

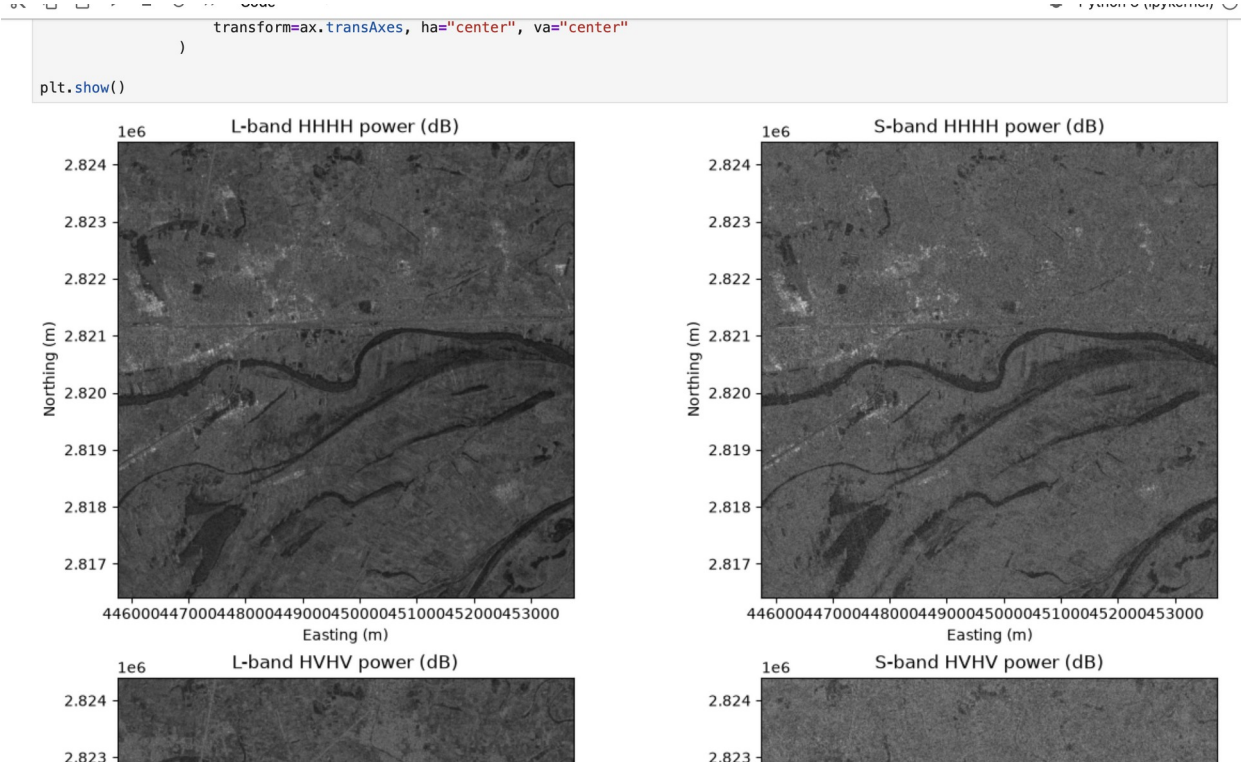
4. Processing completed

4.1 Read the covariance power terms

The real-valued diagonal covariance layers `HHHH` and `HVHV` were extracted as HH and HV power for each frequency. The complex S-band `HHHV` cross-covariance layer was identified but was not used as a direct water-backscatter channel.

4.2 Convert positive power to decibels

For visualization and relative thresholding, positive power values were expressed as 10 log10(power). Invalid pixels were excluded using the GCOV mask. The L-band and S-band HH/HV panels were plotted in projected coordinates.



HH and HV power displayed in decibels for the common analysis window. The dark river corridor is visible in both bands.

4.3 Form a shared analysis window

An 800 × 800 pixel window was extracted around the study centre on a shared 10 m coordinate grid. The notebook reports 640,000 jointly valid pixels. At 100 m² per pixel, the window covers 64 km².

4.4 Build an equal-weight evidence score

The four layers (L-HH, L-HV, S-HH, S-HV) were normalized using the 2nd and 98th percentiles and combined with equal contribution into a relative low-backscatter evidence score. Its reported 10th, 50th and 90th percentiles are 0.263, 0.465 and 0.761.

4.5 Explore local-contrast adaptive weighting

A second score uses local variability/contrast summaries to adapt the contribution of the four layers. The notebook screenshot reports local standard-deviation P50/P90 values by layer and adaptive-score percentiles of 0.243, 0.463 and 0.754. This is a prototype score: the supplied evidence does not show the complete formula, window size or learned/validated weights, so it should not be presented as a validated fusion model.

4.6 Produce threshold candidates and compare masks

Per-band low-HH masks were created using 5th, 10th and 15th percentile cutoffs. A two-band consensus is the intersection of the L-band and S-band low-HH masks at the same nominal percentile. Low-HV conjunctions were also visualized. Eight-connected components were counted for the consensus masks.

5. Exploratory results

5.1 Backscatter distributions

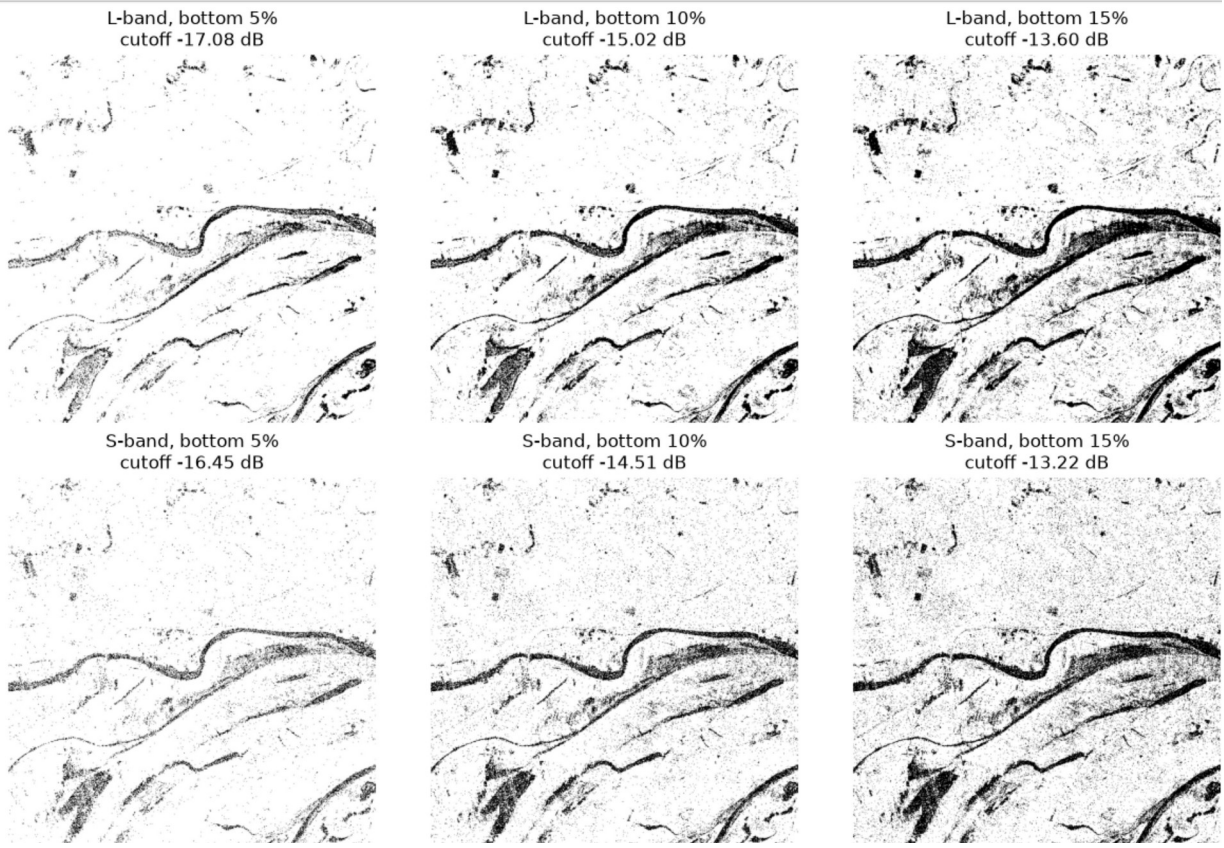
Layer	2nd percentile (dB)	Median (dB)	98th percentile (dB)
L-band HH	−18.41	−9.59	−3.38
L-band HV	−25.89	−18.44	−10.35
S-band HH	−18.33	−8.85	−2.77
S-band HV	−22.78	−15.85	−9.65

The screenshot visualizations show the meandering river corridor and surrounding floodplain texture in both frequency bands. These are visual observations from the plotted window, not a verified delineation of water extent.

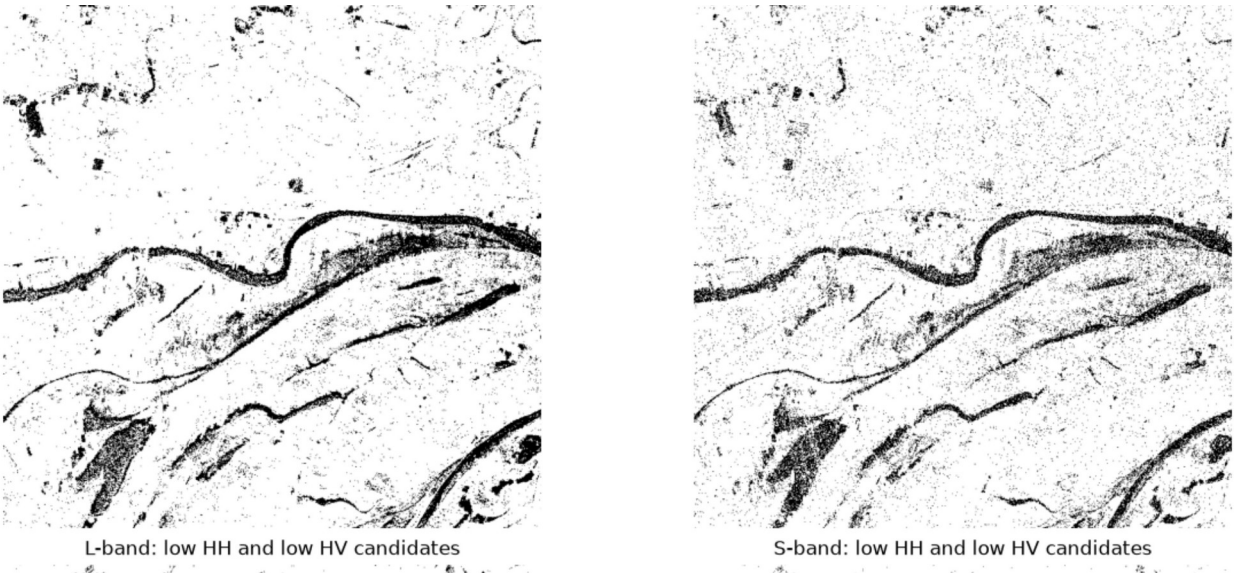
5.2 Threshold sensitivity and cross-band agreement

Nominal percentile	L HH cutoff	S HH cutoff	Consensus pixels	Largest 8-connected patch
5%	−17.08 dB	−16.45 dB	13,783	250 px (2.50 ha)
10%	−15.02 dB	−14.51 dB	35,651	5,524 px (55.24 ha)
15%	−13.60 dB	−13.22 dB	55,475	15,401 px (154.01 ha)

The 10th-percentile band-specific candidate sets contain 64,000 pixels each. Their intersection contains 35,651 pixels and their union contains 92,349 pixels (Jaccard overlap 0.386). Across the 5%, 10% and 15% settings, the reported Jaccard values are 0.274, 0.386 and 0.406. Increasing the threshold increases both the candidate area and the largest connected patch, demonstrating material sensitivity to the chosen cutoff.



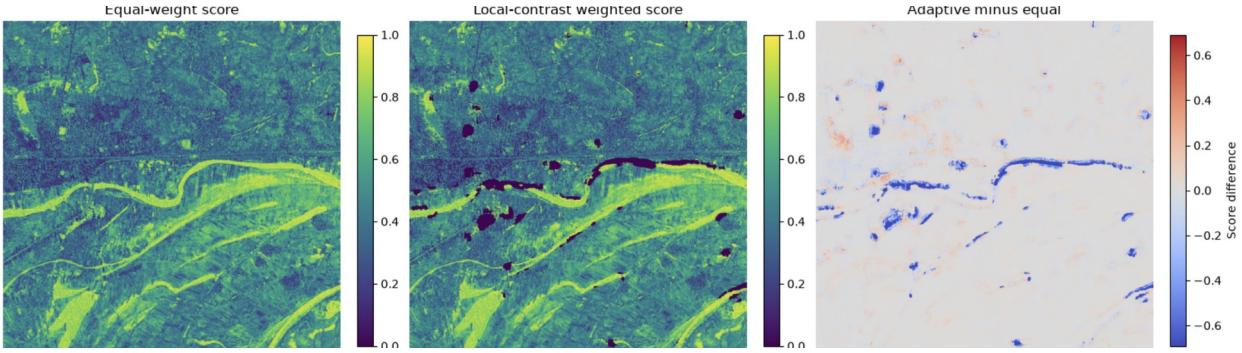
Threshold sensitivity: low-HH candidate masks at the 5th, 10th and 15th percentiles for each band.



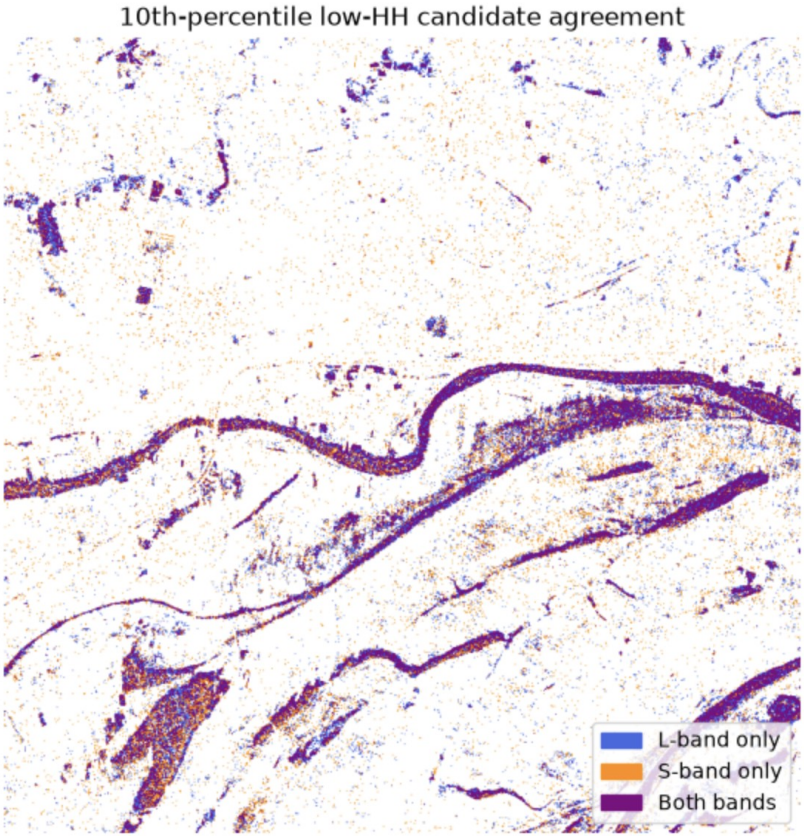
Single-band low-HH and joint low-HH/low-HV candidate previews for L-band and S-band.

5.3 Equal-weight and adaptive-score comparison

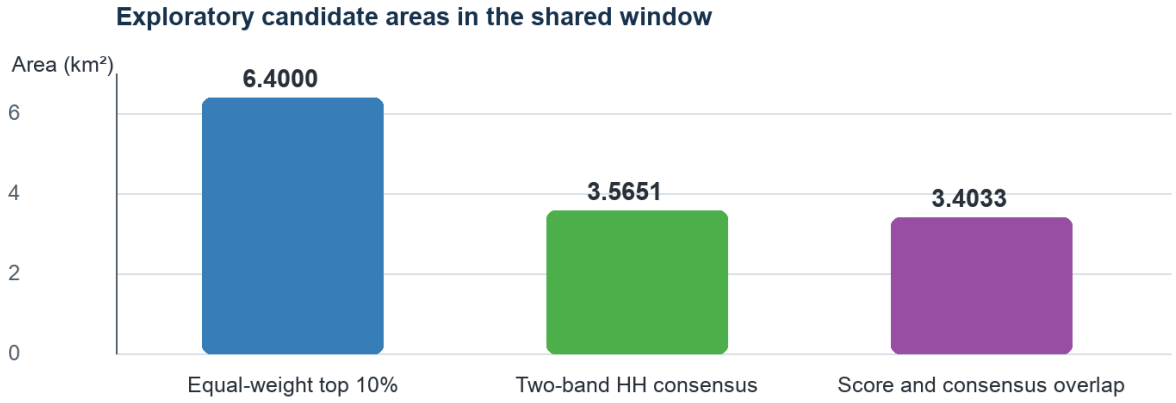
The equal-weight score's top 10% contains 64,000 pixels (6.4000 km²). The 10% two-band HH consensus contains 35,651 pixels (3.5651 km²). Their overlap contains 34,033 pixels (3.4033 km²), with a reported Jaccard overlap of 0.519. The adaptive and equal-weight scores were compared across 640,000 pixels; the notebook reports absolute-difference percentiles (50th, 90th, 99th) of 0.0209, 0.0824 and 0.691, and mean absolute difference 0.0459. This indicates that the adaptive weighting changes some locations appreciably, while most pixelwise differences are more modest.



Equal-weight and local-contrast weighted scores, alongside their pixelwise difference. The weighting result is exploratory.



Spatial agreement between the 10th-percentile L-band and S-band low-HH candidate masks.



Notebook area summary for the equal-weight score, two-band consensus and their overlap.

6. Interpretation and limitations

- The common footprint and mask checks support a like-for-like local comparison of HH/HV backscatter in the selected window.
- Low backscatter and cross-band agreement are useful screening evidence for open water, but they are not sufficient to distinguish water from radar shadow, smooth bare surfaces, or other low-return land covers.
- The reported "candidate area" is thresholded pixel area, not a validated water-area estimate. The 10% consensus area of 3.5651 km² should be read as an exploratory mask area only.
- The adaptive weighting is exploratory. The current evidence does not include a complete reproducible weight formula, independent reference data, or accuracy metrics such as precision, recall, or F1.
- The L-band and S-band acquisitions are near-simultaneous on 30 October 2026. This pair supports a frequency/polarization comparison, not seasonal or temporal change detection. Stable water, new inundation, and recession require multiple dates and consistent classification.
- The notebook's first small-area alignment attempt produced an indexing error; the later shared grid was corrected and reports 800 × 800 arrays with 640,000 valid pixels. The report uses only the corrected shared-window outputs.

7. Next analysis steps

1. Preserve the notebook and record the exact adaptive-weight equation, neighborhood size, per-channel weight maps and all normalization constants.
2. Export the 800 × 800 layers, validity mask, equal-weight score, adaptive score and threshold masks as georeferenced rasters with EPSG:32 645 and the correct 10 m affine transform.
3. Check candidate masks against independent water reference data or manual samples, stratified across river, floodplain, agriculture and built-up classes.
4. Run connected-component filtering and assess whether candidate patches follow plausible channel and floodplain geometry; report sensitivity across cutoffs.
5. Acquire multiple dates in relevant seasons, harmonize masks and thresholds, then derive stable water, water gain and water loss only after per-date classification and validation.
6. Compare L-band and S-band performance separately and together; report metrics by polarization and land-cover context before selecting a final fusion method.

References

[1] NASA. NISAR Mission Overview. https://science.nasa.gov/mission/nisar/mission-overview/ (accessed 10 November 2026).

[2] Alaska Satellite Facility DAAC. NISAR Data User Guide: Geocoded Polarimetric Covariance (GCOV). https://nisar-docs.asf.alaska.edu/gcov/ (accessed 10 November 2026).

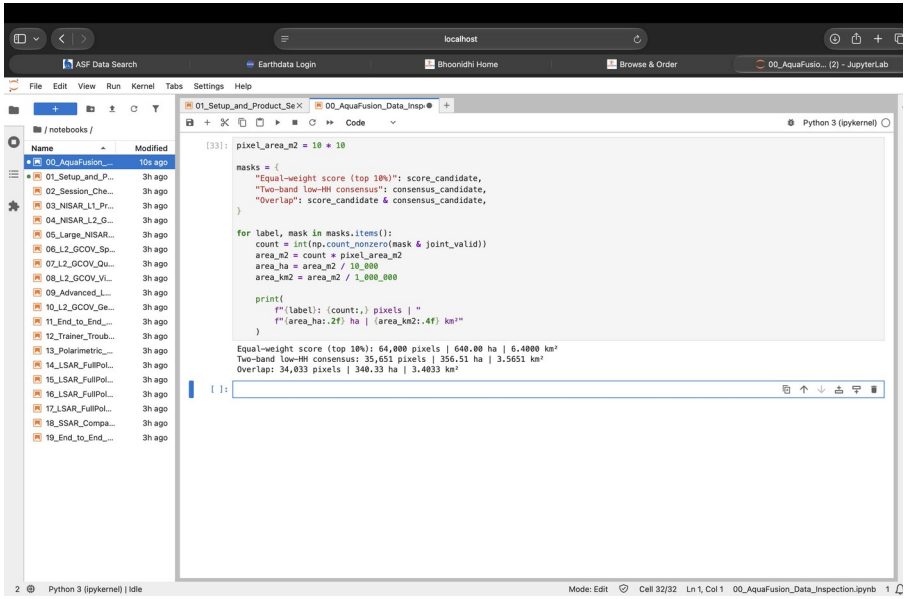
[3] Indian Space Research Organisation, Space Applications Centre. NISAR L & S Band Level-1 & Level-2 Data Product Format Document, Version 1.2.1. QA report supplied with the S-band product: NISAR_S2_UR_GCOV_034_098_A_014_3700_DHNA_A_20 261 030T233 811_20 261 030T233 847_P00 500_M_F_I_001 _QA.pdf.

[4] ASF DAAC Vertex. NISAR product search and scene metadata, screenshots supplied by the project team, 10 November 2026.

[5] AquaFusion NISAR data-inspection notebook outputs, screenshots supplied by the project team, 10 November 2026.

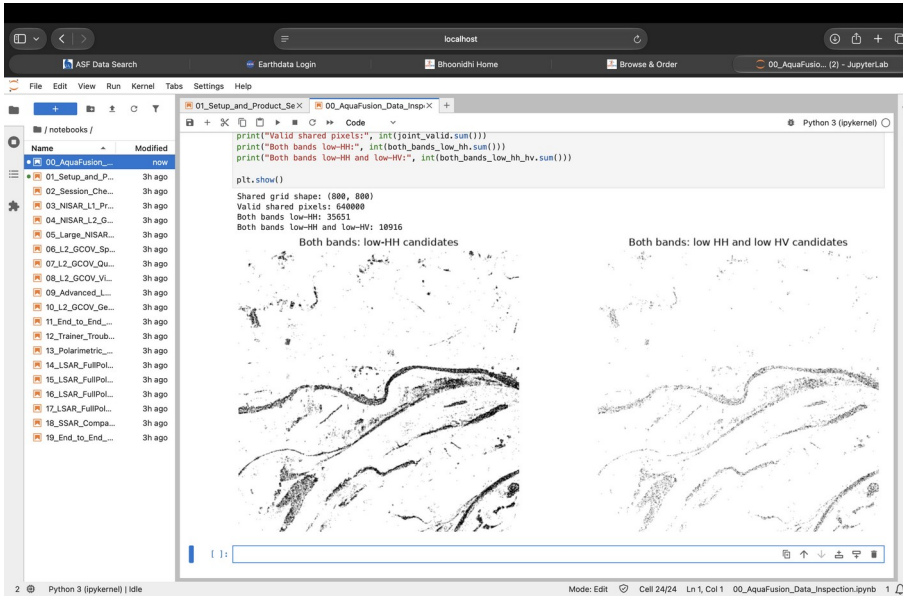
Appendix A Supplied pictures

This appendix retains all 30 picture files listed with the request, in their supplied order. Captions describe visible content and distinguish notebook analysis from setup and search context. Some screenshots show repeated views of the same result; repeats are retained as requested.



Picture 01. Candidate-area counts and estimated area for the equal-weight score, two-band consensus and overlap.

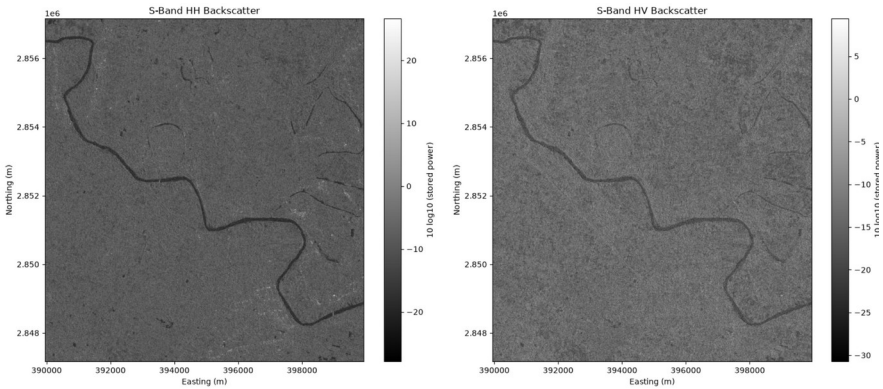
codex-clipboard-6 836 806e-4361-466a-a172-72 843cba5e28.png



Picture 02. Cross-polarization candidate masks and the shared 800 by 800 pixel window.

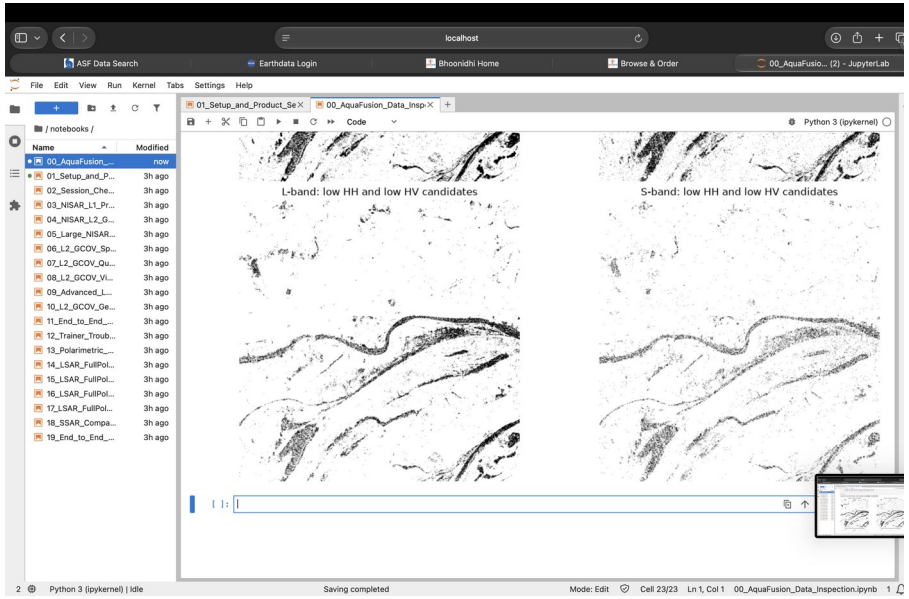
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APPENDIX | ALL SUPPLIED PICTURES



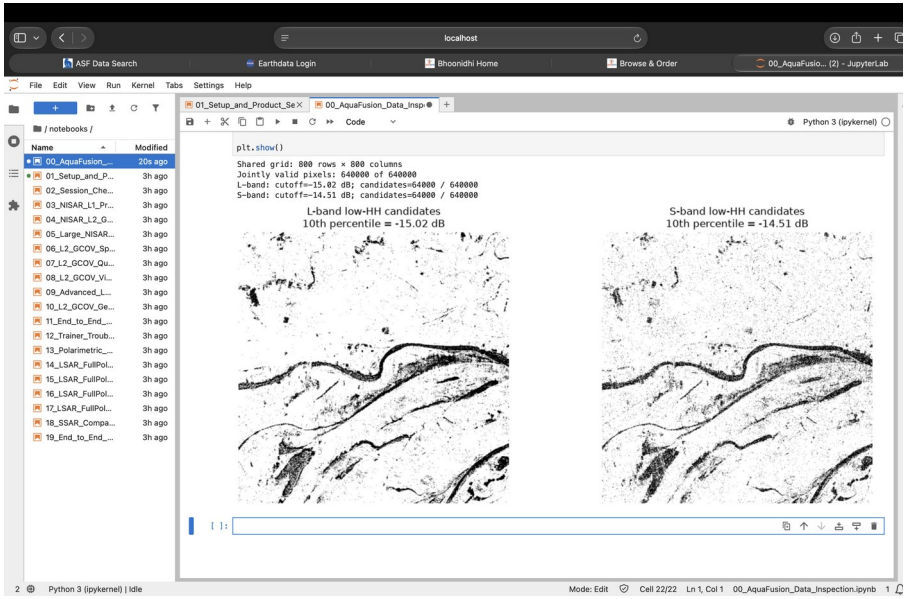
Picture 03. S-band HH and HV backscatter panels with map coordinates.

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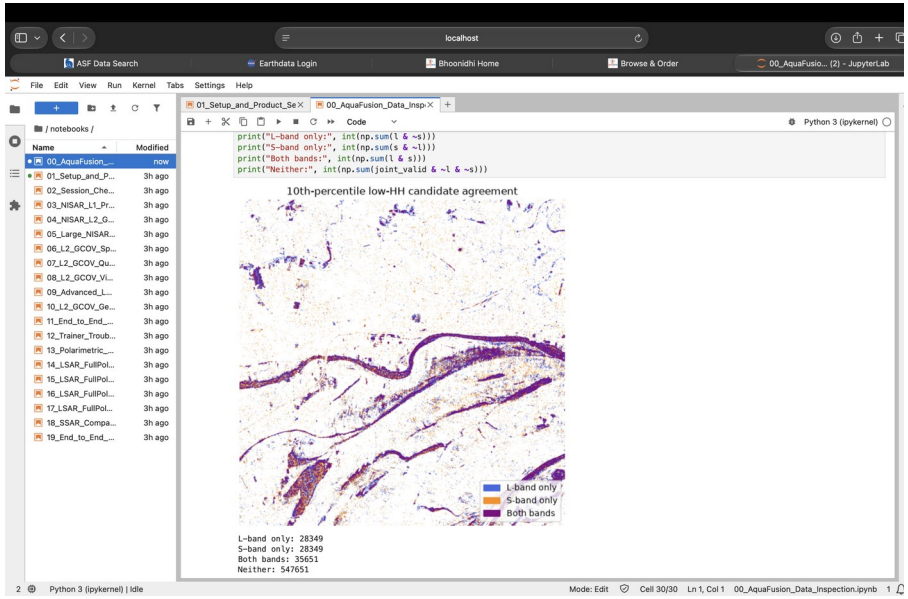
Picture 04. Low-HH and low-HH/low-HV candidate previews for L-band and S-band.

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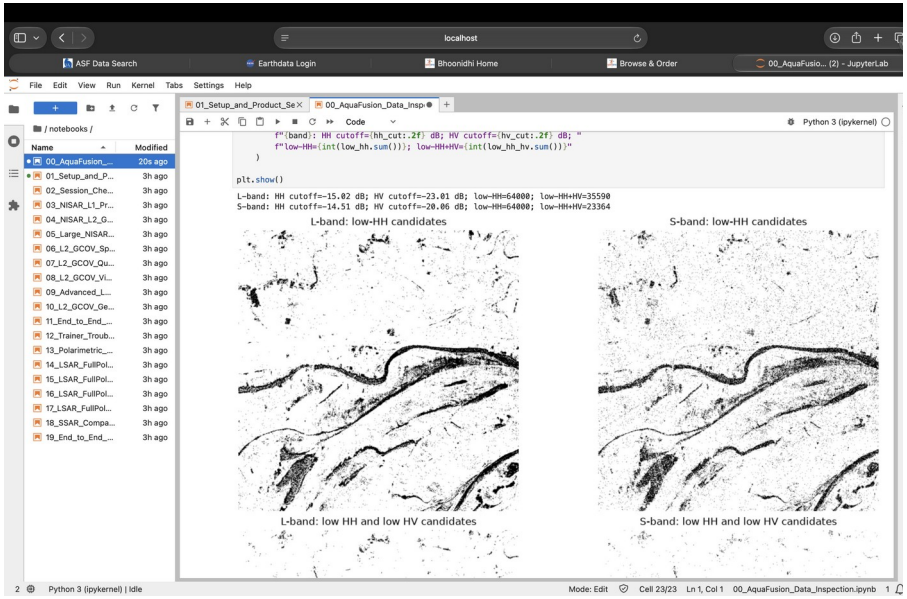
Picture 05. JupyterLab project workspace and notebook setup.

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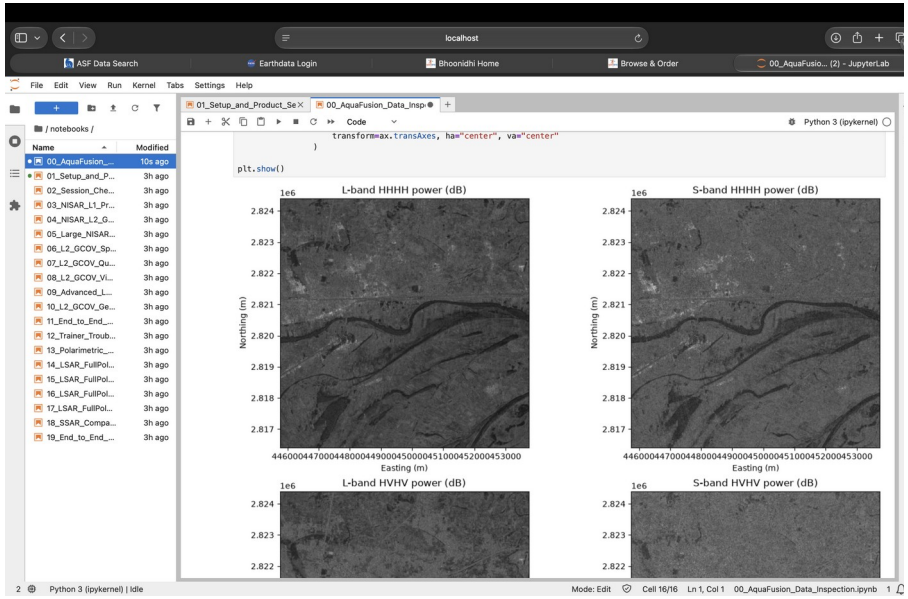
Picture 06. Candidate maps at the 5th, 10th and 15th percentile thresholds.

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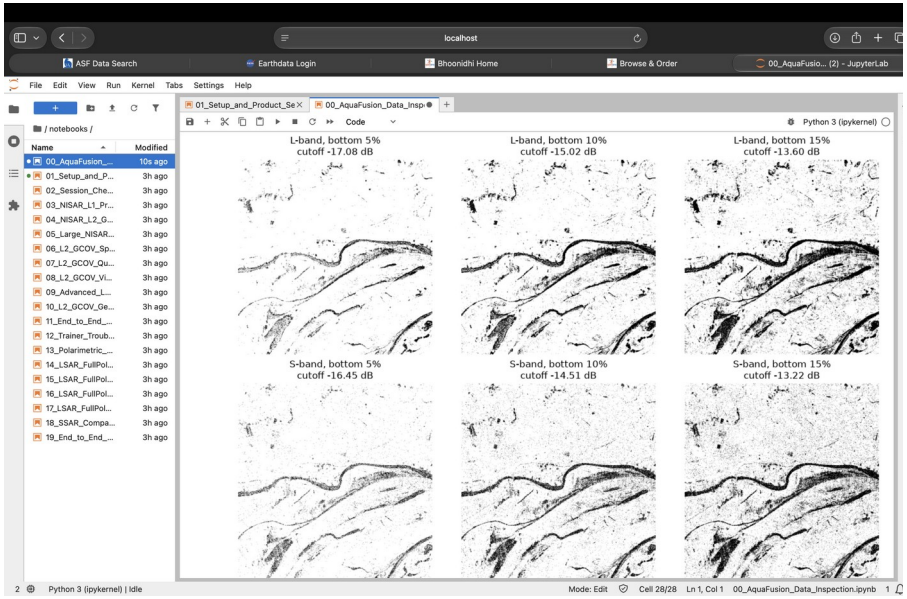
Picture 07. Per-band low-HH candidate maps and two-polarization candidate masks.

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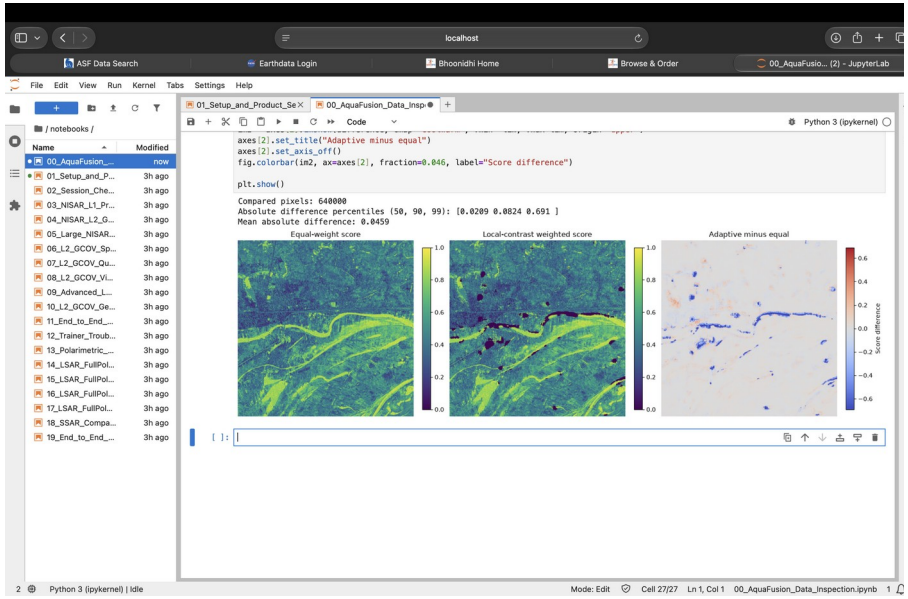
Picture 08. HH and HV power visualization for L-band and S-band.

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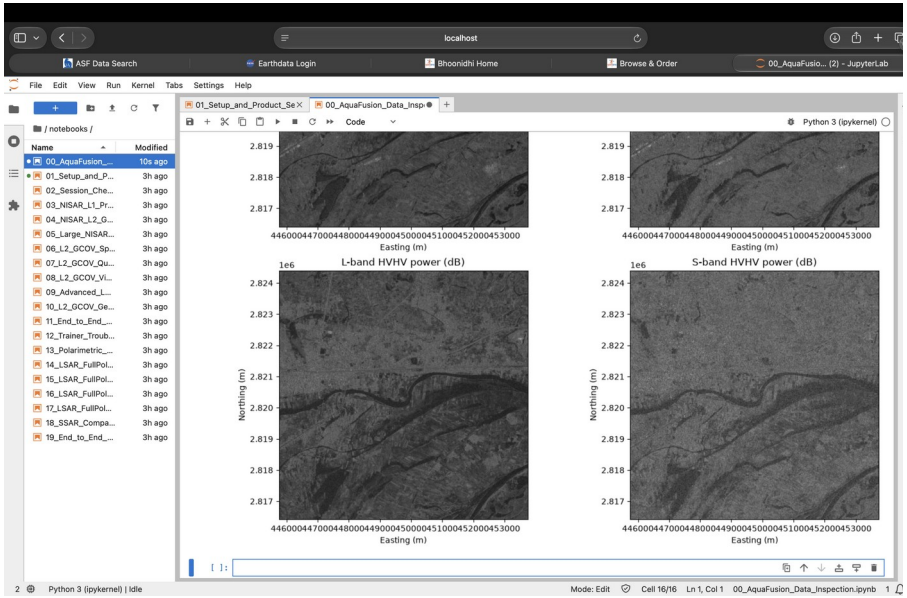
Picture 09. Sensitivity of low-HH candidates to alternative percentile cutoffs.

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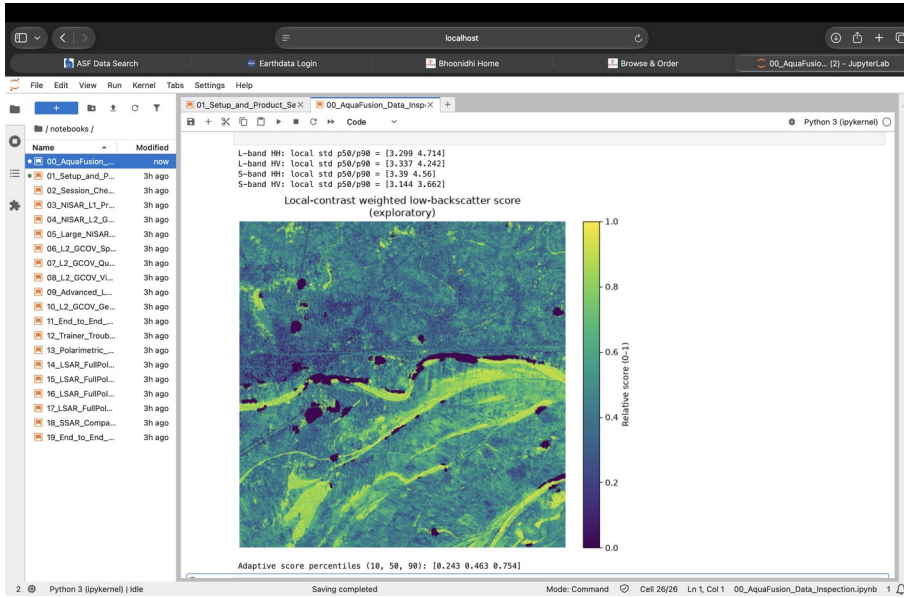
Picture 10. Comparison of equal-weight and local-contrast adaptive scores, with their difference.

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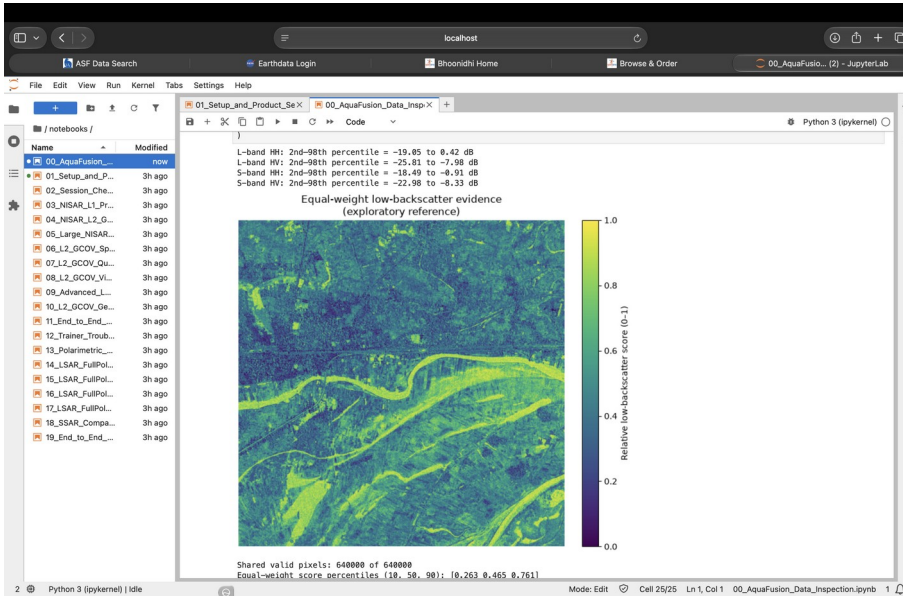
Picture 11. Local-contrast weighted low-backscatter score and reported local variability summaries.

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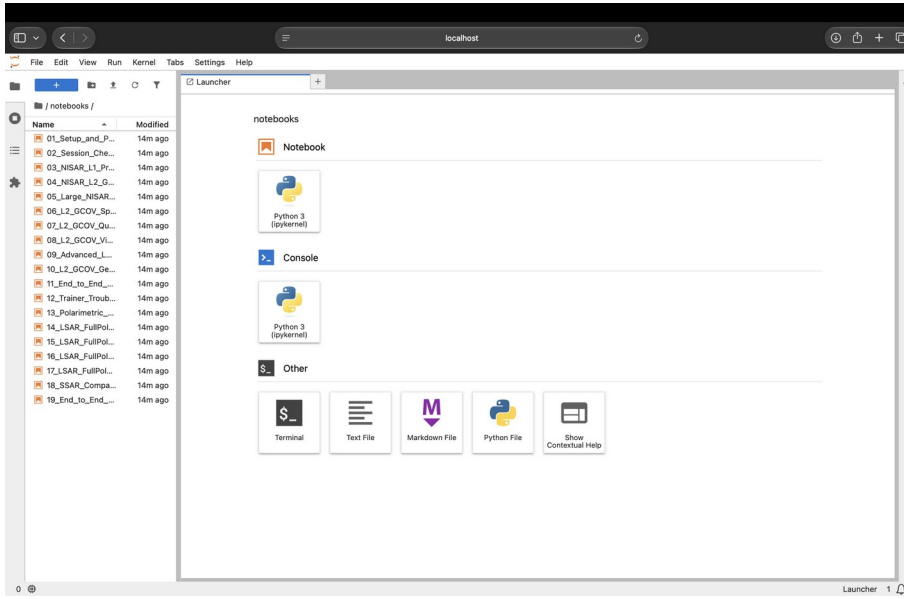
Picture 12. Equal-weight low-backscatter evidence score and score percentiles.

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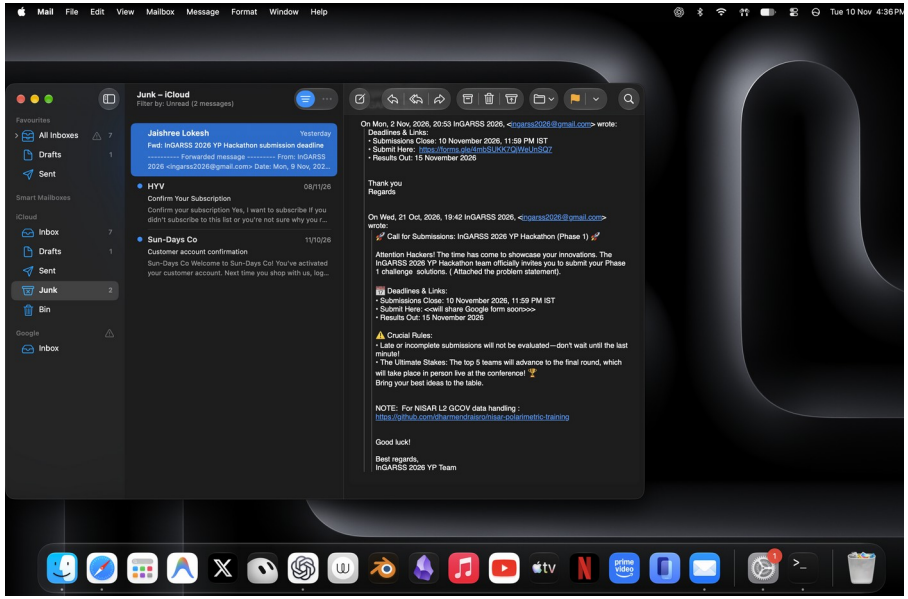
Picture 13. Percentile thresholds and connected-component summaries for HH consensus.

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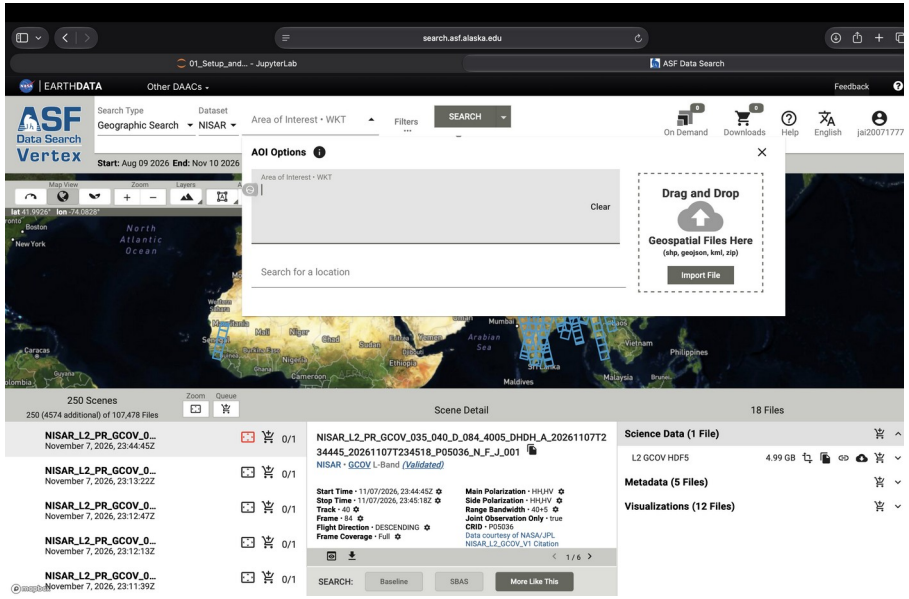
Picture 14. JupyterLab launcher and project notebook list.

Screenshot 2026-11-10 at 4.29.21 PM.png



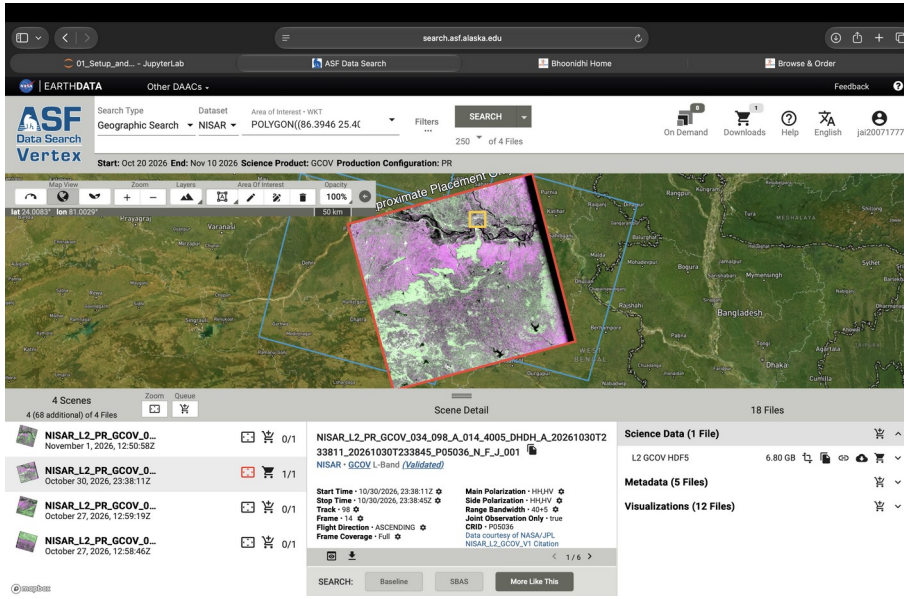
Picture 15. Email screenshot showing the submission notice and external deadline context.

Screenshot 2026-11-10 at 4.36.48 PM.png



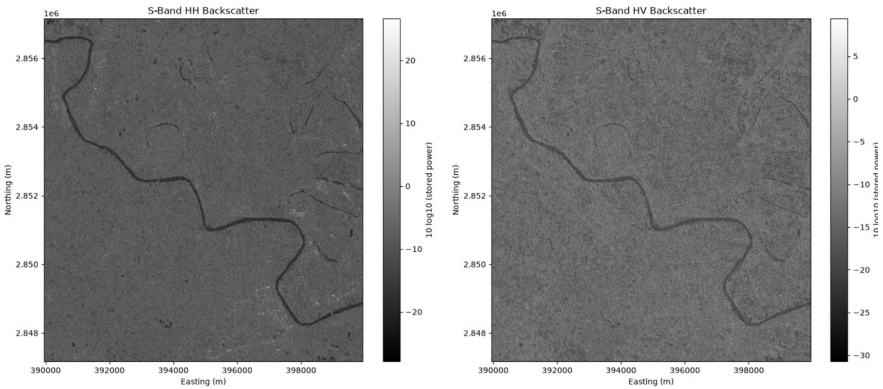
Picture 16. ASF Vertex search interface and NISAR product listing.

Screenshot 2026-11-10 at 4.54.29 PM.png



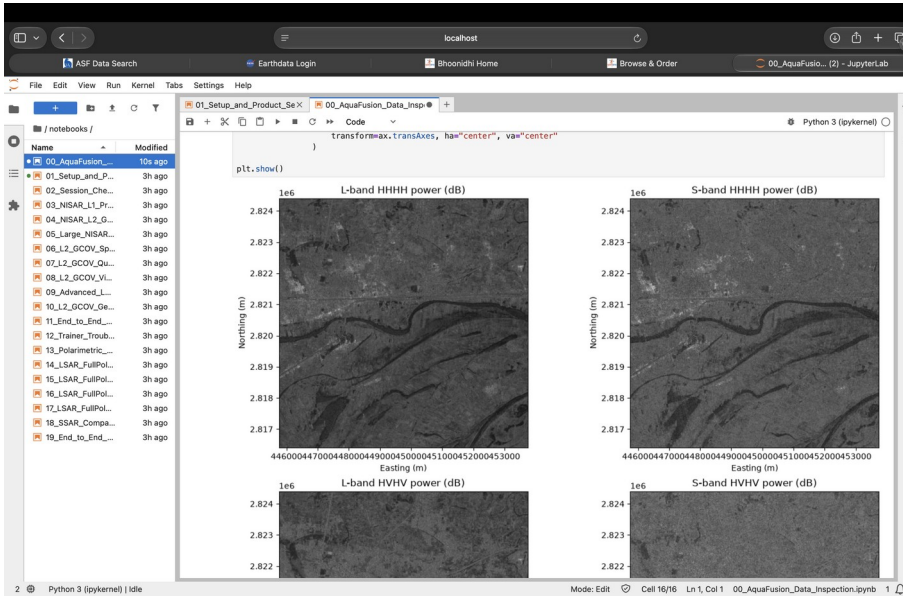
Picture 17. ASF Vertex scene selection showing the Bihar-area scene footprint and product metadata.

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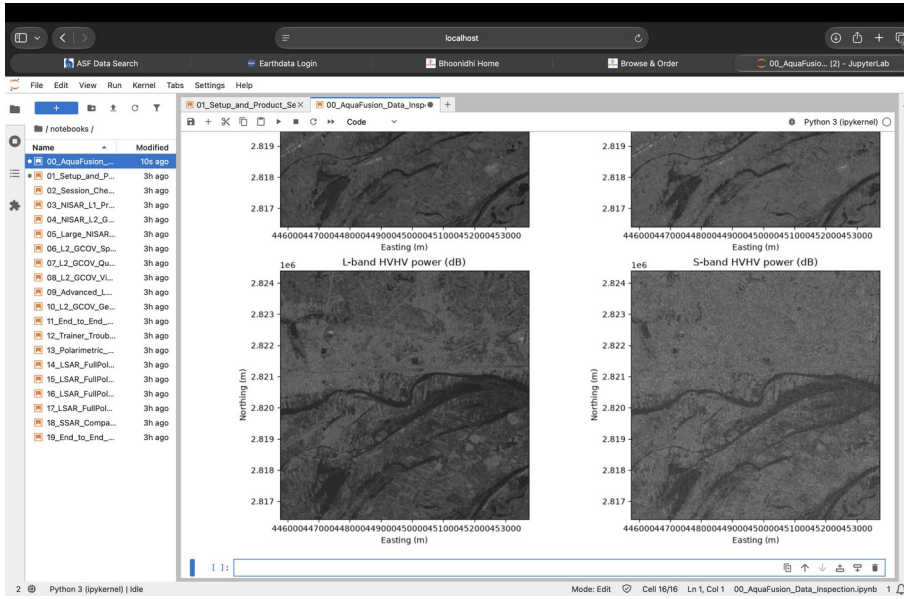
Picture 18. S-band HH and HV backscatter visualization.

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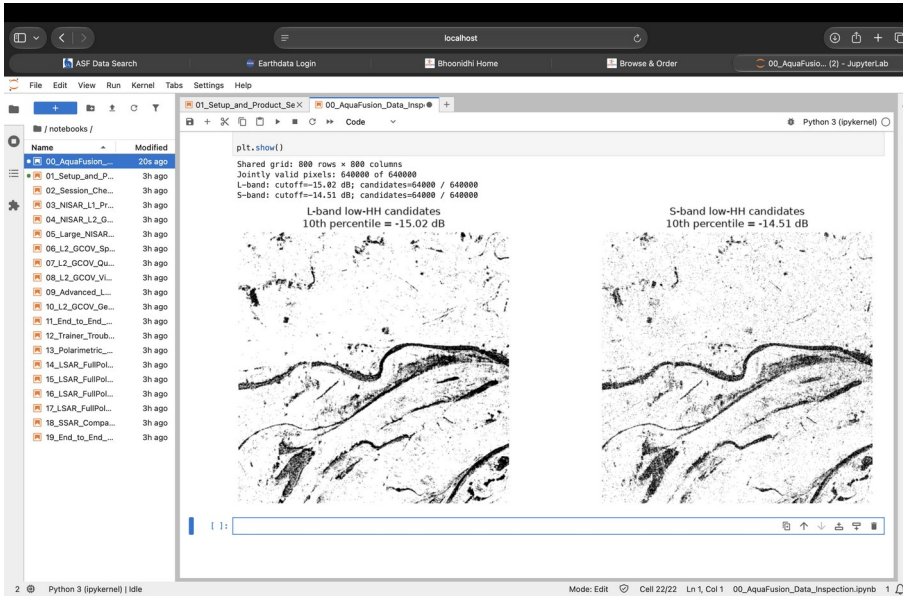
Picture 19. Four-panel L-band and S-band HH/HV power visualization.

Screenshot 2026-11-10 at 9.18.13 PM.png



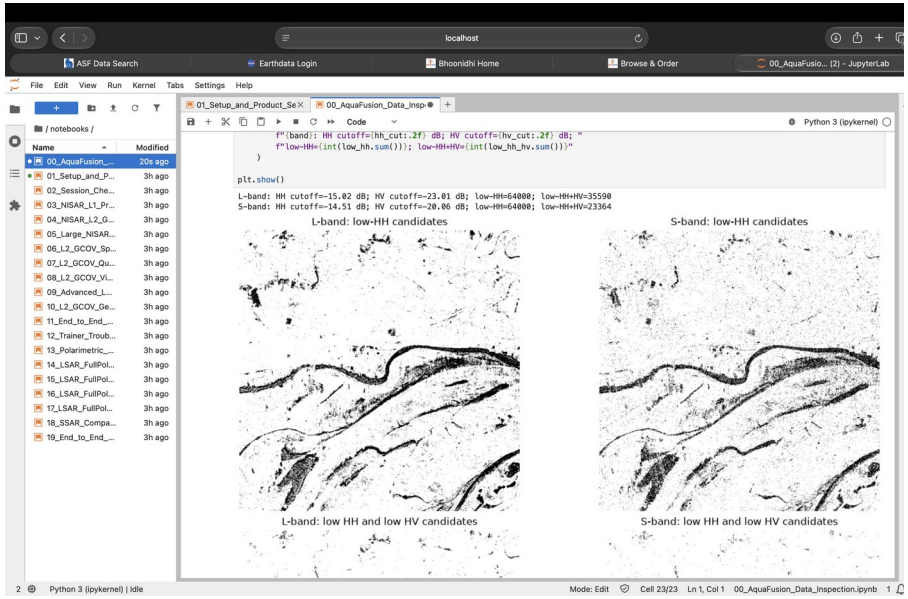
Picture 20. Lower portion of the four-panel L-band and S-band HH/HV visualization.

Screenshot 2026-11-10 at 9.18.21 PM.png



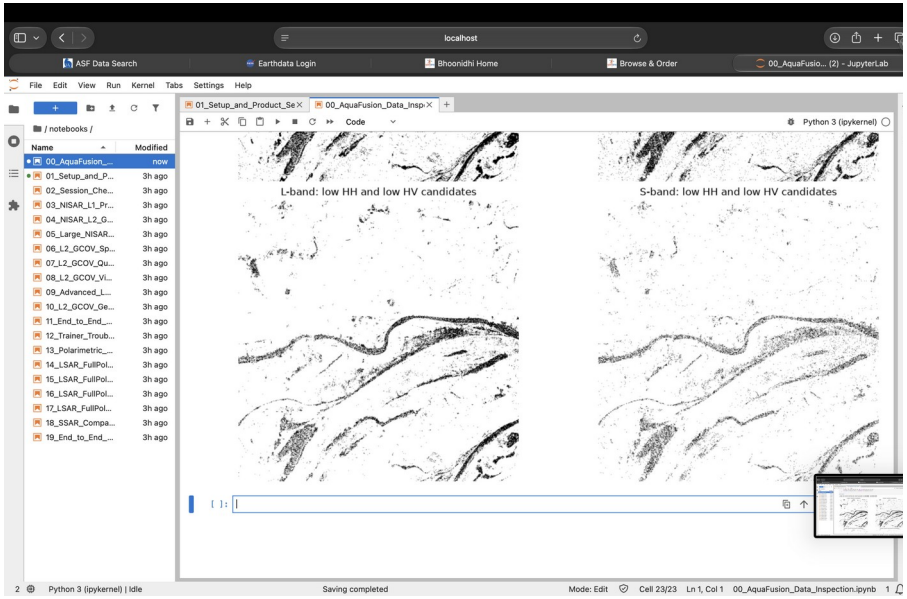
Picture 21. L-band and S-band exploratory low-HH candidate previews.

Screenshot 2026-11-10 at 9.23.57 PM.png



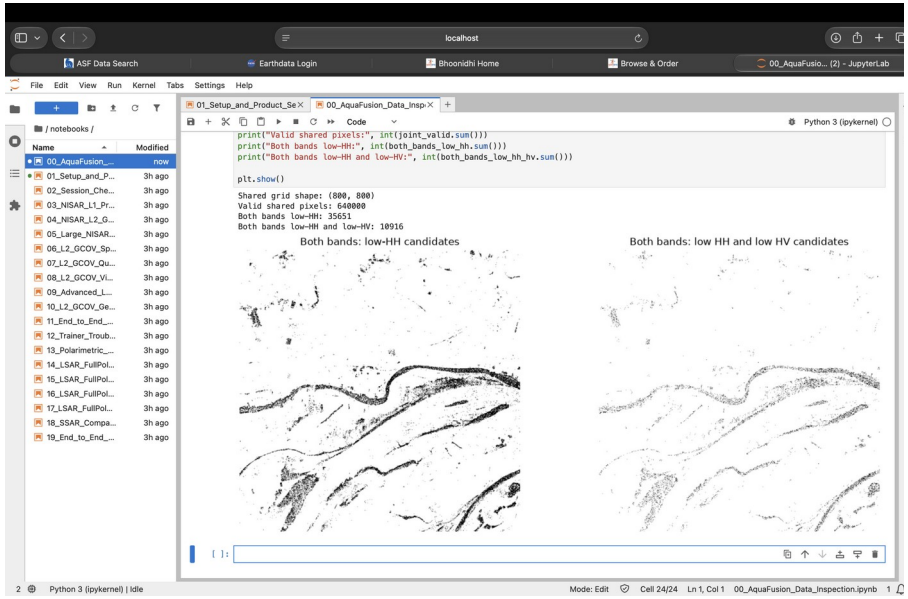
Picture 22. Low-HH and low-HV candidate previews for both radar bands.

Screenshot 2026-11-10 at 9.24.45 PM.png



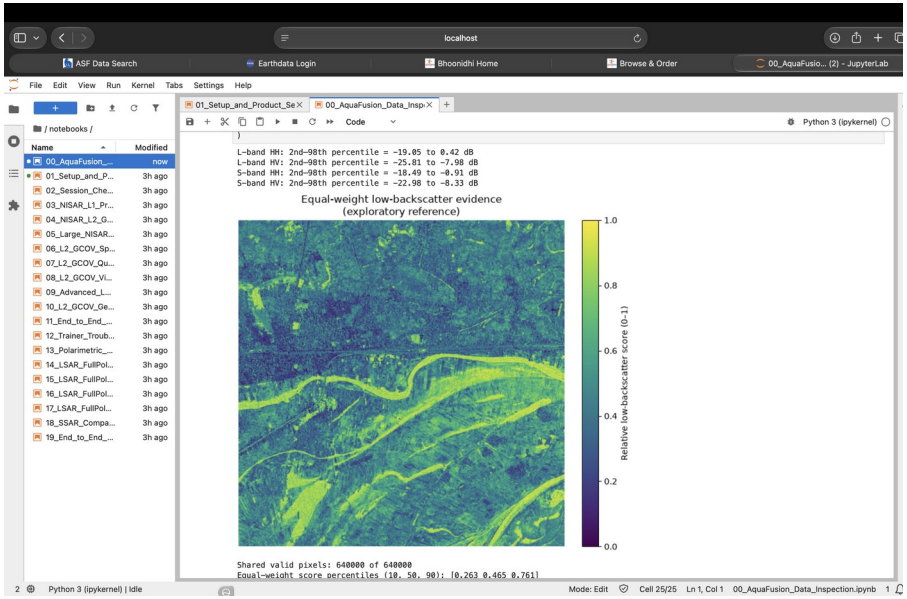
Picture 23. Repeated view of the two-polarization candidate previews.

Screenshot 2026-11-10 at 9.24.48 PM.png



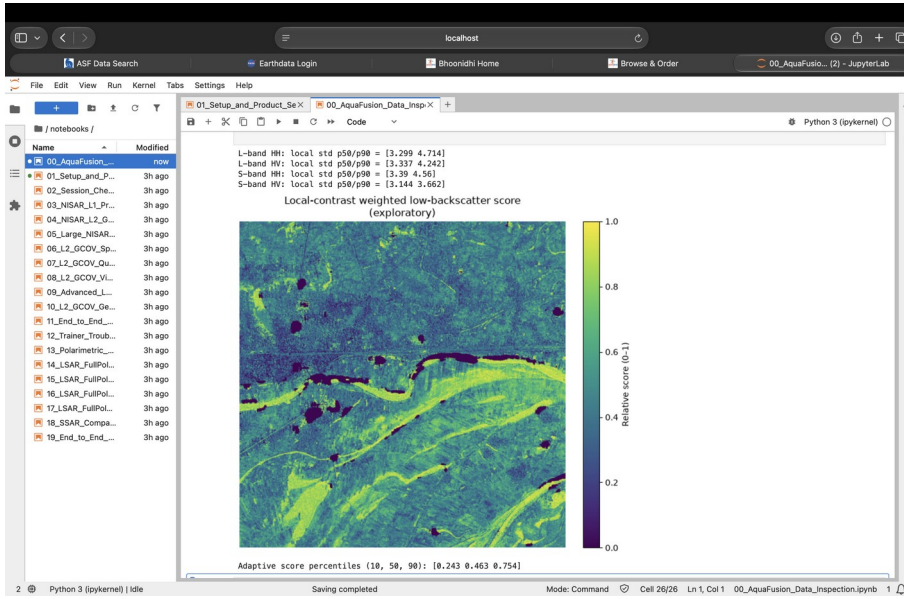
Picture 24. Intersection of low-HH candidate pixels across bands.

Screenshot 2026-11-10 at 9.26.55 PM.png



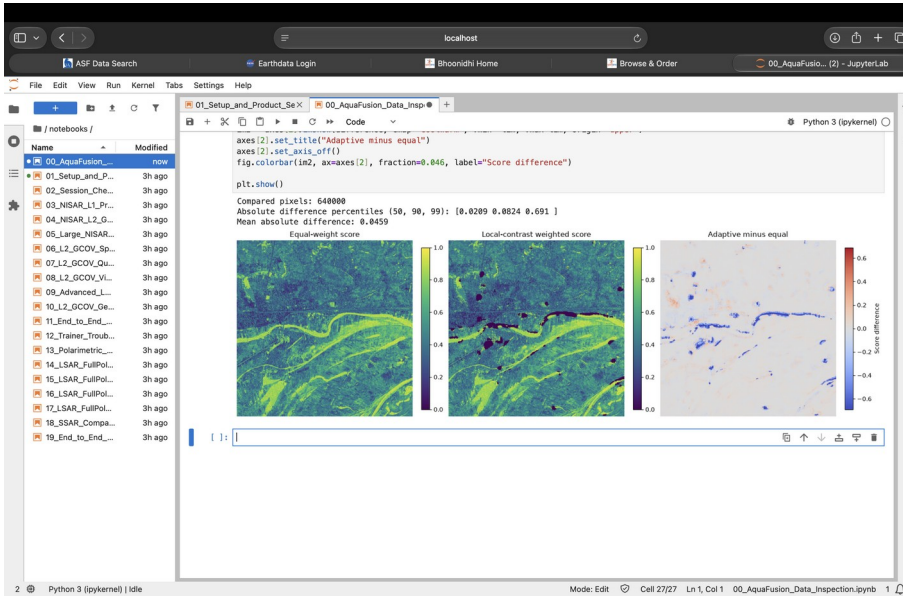
Picture 25. Equal-weight low-backscatter evidence score.

Screenshot 2026-11-10 at 9.28.54 PM.png



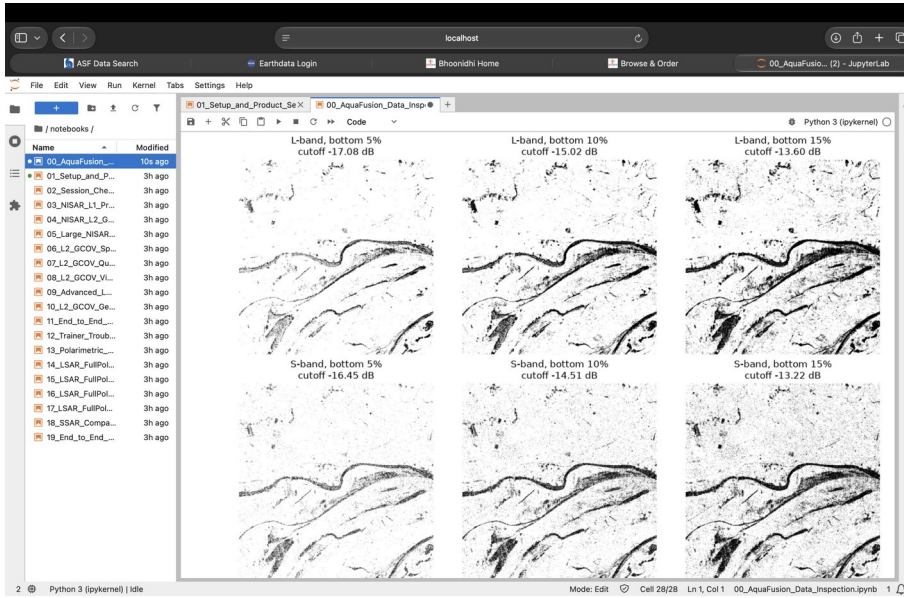
Picture 26. Exploratory local-contrast weighted score.

Screenshot 2026-11-10 at 9.30.48 PM.png



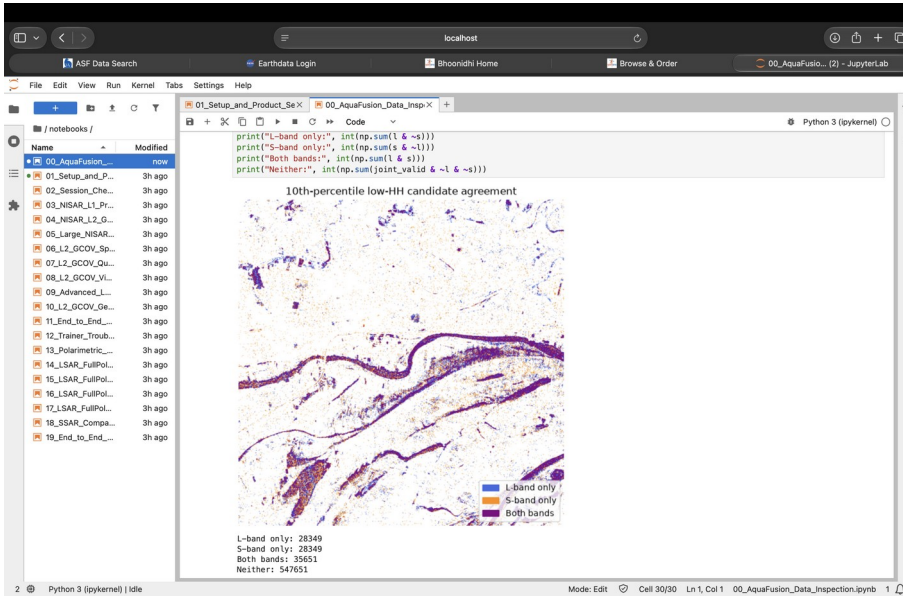
Picture 27. Equal-weight score, adaptive score and difference comparison.

Screenshot 2026-11-10 at 9.31.27 PM.png



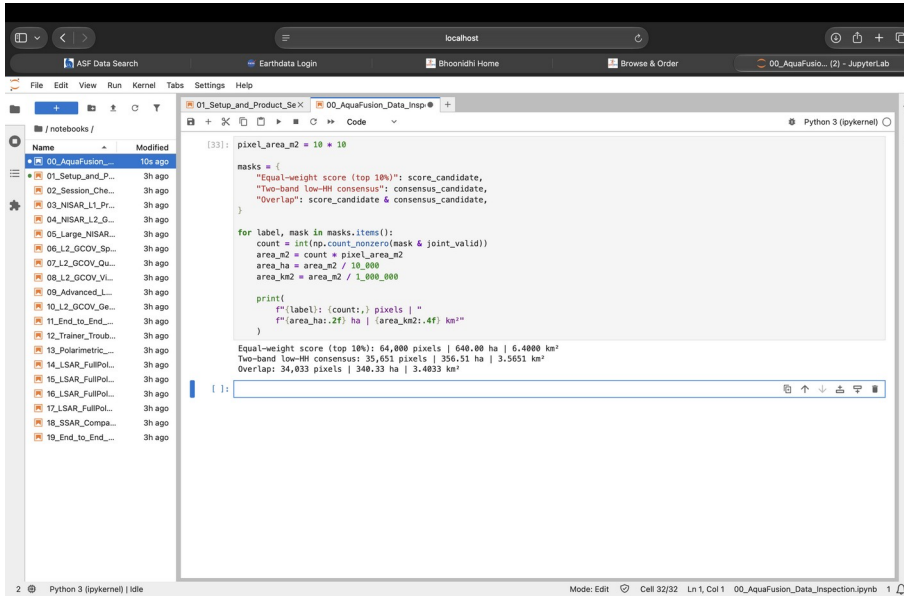
Picture 28. Threshold sensitivity maps at the 5th, 10th and 15th percentiles.

Screenshot 2026-11-10 at 9.33.15 PM.png



Picture 29. Spatial agreement between the L-band and S-band low-HH candidate masks.

Screenshot 2026-11-10 at 9.46.28 PM.png



Picture 30. Estimated area for score, consensus and overlap candidate masks.

Screenshot 2026-11-10 at 9.48.17 PM.png